



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS
BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONOURS

CSE3433 SOFTWARE ARCHITECTURE

SEMESTER II SESSION 25/26

PROJECT TITLE:
GROUP 14
RECRUITX - EVENT-DRIVEN JOB RECRUITMENT & APPLICATION PROCESSING
SYSTEM

INDIVIDUAL LOGBOOK

PREPARED FOR:
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PREPARED BY:

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1. Individual Logbook Report	3
1.1. Team Information	3
1.2. Team Members	3
2. Weekly Progress Log	4
2.1. Week 4: Requirement Analysis and Service Identification	4
2.2. Week 5: Functional Requirements and Architecture Planning	4
2.3. Week 6: Architecture Design and Diagram Development	5
2.4. Week 7: SOA Migration Documentation and Service Boundary Definition	5
2.5. Week 8: Authentication Flow and API Gateway Design	6
2.6. Week 9: Ticket Service Backend Development	7
2.7. Week 10: Business Logic and Security Enhancement	7
2.8. Week 11: Service Integration and Deployment Preparation	8
2.9. Week 12: Prototype Validation and Final Documentation	8
3. GenAI Usage Record	9
3.1. Reflection	11
4. Evidence Summary	11
5. Final Individual Reflection	12

1. Individual Logbook Report

RecruitX - Event-Driven Job Recruitment & Application Processing System

1.1. Team Information

Item	Details
Name	Nur Alin Hayani Binti Ahmad Syukri
Matric No	S75034
Course	Bachelor of Computer Science (Software Engineering) with Honours
Subject	CSE3433 Software Architecture
Project Title	
Group Number	Group 14
Architecture Style	Event-Driven Architecture (EDA)

1.2. Team Members

Name	Matric No	Role
Nur Alin Hayani Binti Ahmad Syukri	S75034	Candidate Portal Development, Application Processing, Database Integration
Ainaa Nadhirah Binti Asmadi	S75246	Recruiter Portal Development, Candidate Management
Maryam Musfirah Binti Ariff Hazizan	S75922	Interview Scheduling, Notification Service, Documentation

2. Weekly Progress Log

2.1. Week 4: Requirement Analysis and Service Identification

Category	Details
Activities	• Discussed project scope and selected RecruitX as the project topic. • Analysed problems in traditional recruitment processes. • Identified candidate and recruiter user roles. • Identified key events required in the system.
Individual Contributions	• Proposed Event-Driven Architecture (EDA) as the architecture style. • Defined core recruitment events: ◦ JobPosted ◦ ApplicationReceived ◦ ApplicationStatusUpdated ◦ InterviewScheduled
Challenges	• Understanding how recruitment workflows could be transformed into event-based processes.
Action Plan	• Study publish-subscribe communication model and EDA concepts.

2.2. Week 5: Functional Requirements and Architecture Planning

Category	Details
Activities	• Defined functional and non-functional requirements. • Planned system modules and architecture structure.
Individual Contributions	• Designed candidate application workflow. • Prepared system requirements documentation. • Identified event producers and consumers.

Challenges	Mapping business processes into asynchronous event flows.
Action Plan	Develop architecture diagrams.

2.3. Week 6: Architecture Design and Diagram Development

Category	Details
Activities	- Developed architecture diagrams. - Prepared architecture design documentation.
Individual Contributions	Designed: - System Context Diagram - Component Architecture Diagram - Interaction Diagram
Challenges	Representing event communication clearly within diagrams.
Action Plan	Start prototype development.

2.4. Week 7: SOA Migration Documentation and Service Boundary Definition

Category	Details
Activities	- Designed Event Bus structure. - Prepared database schema.

Individual Contributions	Created database tables: • users • jobs • applications • event_logs • interviews • notifications • screening_results
Challenges	• Designing database relationships for recruitment workflows.
Action Plan	• Develop a candidate portal.

2.5. Week 8: Authentication Flow and API Gateway Design

Category	Details
Activities	• Developed candidate user interface. • Implemented login authentication.
Individual Contributions	• Built Candidate Portal. • Implemented candidate profile management. • Connected user authentication with database.
Challenges	• Managing session handling and user access control.
Action Plan	• Implement application submission workflow.

2.6. Week 9: Ticket Service Backend Development

Category	Details
Activities	• Developed application submission functionality. • Implemented application processing service.
Individual Contributions	• Developed ApplicationConsumer. • Connected application submissions with database storage. • Implemented ApplicationReceived event processing.
Challenges	• Ensuring application records were created correctly.
Action Plan	• Develop Resume Screening Service.

2.7. Week 10: Business Logic and Security Enhancement

Category	Details
Activities	• Implemented automated screening mechanism.
Individual Contributions	• Developed ResumeScreeningConsumer. • Implemented fit score evaluation logic. • Created screening_results database integration.
Challenges	• Designing meaningful screening criteria.
Action Plan	• Develop recruiter management features.

2.8. Week 11: Service Integration and Deployment Preparation

Category	Details
Activities	• Integrated candidate and recruiter modules. • Tested end-to-end workflows.
Individual Contributions	Connected: • ApplicationConsumer • ResumeScreeningConsumer • NotificationConsumer Verified event logging functionality.
Challenges	• Ensuring asynchronous event processing operated correctly.
Action Plan	• Complete interview scheduling module.

2.9. Week 12: Prototype Validation and Final Documentation

Category	Details
Activities	• Performed system testing. • Validated Event-Driven Architecture workflow. • Completed report documentation.
Individual Contributions	Tested: • Job Posting • Application Submission • Resume Screening • Status Updates

	• Interview Scheduling • Notifications Verified database integration.
Challenges	• RabbitMQ installation issues. • Adopted SimulatedRabbitMQ implementation.
Action Plan	• Finalize project report and presentation materials.

3. GenAI Usage Record

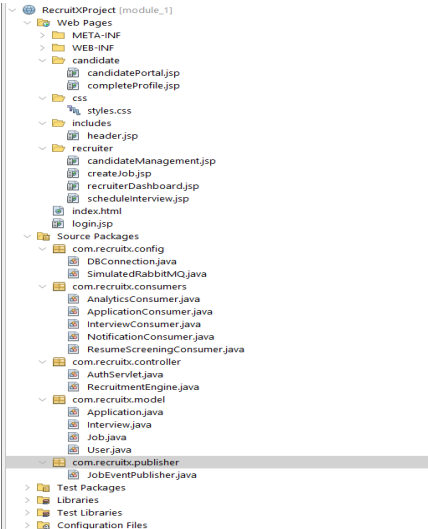
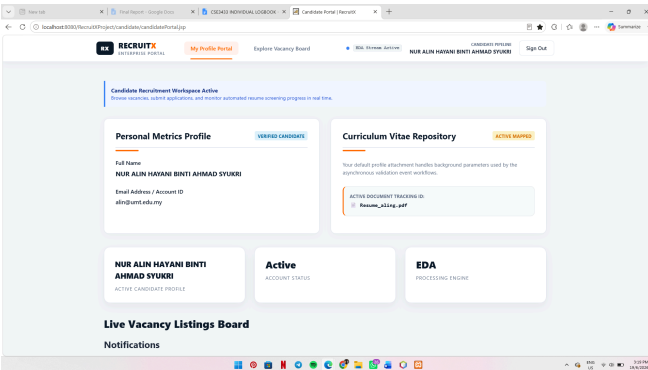
Week	GenAI Tool	Purpose	Prompt Used	Output Summary	Reflection and Adaptation
Week 4	ChatGPT	Assist in understanding Event-Driven Architecture (EDA) concepts and workflows.	"Explain Event-Driven Architecture and how it can be applied to a recruitment management system."	Provided explanation of EDA concepts, event producers, event consumers, publish-subscribe communication, and recruitment workflow examples.	The information was compared with lecture materials and used to justify the selection of EDA as the project's architecture style.
Week 5	ChatGPT	Support requirements analysis and event identification.	"Suggest functional requirements and events for a recruitment system using EDA."	Generated examples of recruitment-related events such as JobPosted, ApplicationReceived, InterviewScheduled, and NotificationGenerated.	The suggestions were reviewed and adapted to match the project scope and business requirements defined by the team.
Week	ChatGPT	Assist in	"Explain the	Produced	The output was used as

k 6		architecture design and diagram planning.	components of an EDA recruitment system and the relationships between services."	descriptions of producers, consumers, event bus mechanisms, and service interactions.	guidance when preparing architecture diagrams and documenting component responsibilities.
Week 8	ChatGPT	Support implementation of candidate application workflow.	"How can application submissions trigger asynchronous processing using event consumers?"	Provided examples of event publishing and asynchronous processing flow.	The concepts were adapted into the SimulatedRabbitMQ implementation used within RecruitX.
Week 9	ChatGPT	Assist in database integration and service implementation.	"Suggest database tables for a recruitment system with event logging."	Generated recommendations for applications, event_logs, notifications, interviews, and screening_results tables.	The suggestions were reviewed and modified to meet project requirements before implementation.
Week 10	ChatGPT	Assist in designing resume screening logic.	"How can a resume screening service calculate a fit score?"	Suggested keyword-based screening methods and scoring approaches.	The team implemented a simplified fit score mechanism suitable for demonstrating automated screening functionality.
Week 11	ChatGPT	Support debugging and service integration.	"Review EDA workflow and identify missing consumer interactions."	Provided recommendations for improving event flow and consumer communication.	The outputs were used to verify implementation consistency and improve system integration.
Week 12	ChatGPT	Assist in final report preparation and documentation.	"Generate report sections for architecture evaluation and implementation."	Produced report drafts, explanations, and documentation structure suggestions.	All generated content was edited, verified, and rewritten where necessary before inclusion in the final report and logbook.

3.1. Reflection

GenAI served as a supplementary learning and documentation tool throughout the development of RecruitX. It helped improve understanding of Event-Driven Architecture, provided guidance during system design discussions, and assisted in organizing project documentation. However, all technical decisions, coding activities, testing procedures, and implementation outcomes remained the responsibility of the project team. The use of GenAI improved productivity while maintaining academic integrity through continuous validation and adaptation of generated outputs.

4. Evidence Summary

Evidence	Description
	The project structure demonstrates the separation of responsibilities through controllers, consumers, publishers, models, and configuration classes. This structure supports maintainability and modular development.
	The Candidate Portal allows users to view available vacancies, manage their profiles, monitor application progress, and receive recruitment updates.

The Recruiter Dashboard provides access to vacancy management, candidate management, interview scheduling, and event monitoring functions.

		id	event_type	payload	processed_at		
☐	Edit	☐	Copy	Delete	1 ApplicationReceived	jobId:1,candidateId:3,resume:Resume_aling.pdf	2026-06-07 15:57:17
☐	Edit	☐	Copy	Delete	2 ApplicationReceived	jobId:1,candidateId:3,resume:Resume_aling.pdf	2026-06-09 03:18:24
☐	Edit	☐	Copy	Delete	3 ApplicationStatusUpdated	applicationId:1,status:ACCEPTED,recruiterId:1	2026-06-09 03:28:35
☐	Edit	☐	Copy	Delete	4 ApplicationStatusUpdated	applicationId:2,status:SHORTLISTED,recruiterId:1	2026-06-09 03:35:37
☐	Edit	☐	Copy	Delete	5 ApplicationStatusUpdated	applicationId:2,status:INTERVIEW,recruiterId:1	2026-06-09 03:50:01
☐	Edit	☐	Copy	Delete	6 ApplicationReceived	jobId:1,candidateId:6,resume:Resume_afiq.pdf	2026-06-09 14:45:10
☐	Edit	☐	Copy	Delete	7 JobPosted	title:lead backed software engineer	2026-06-09 15:05:29
☐	Edit	☐	Copy	Delete	8 ApplicationReceived	jobId:1,candidateId:6,resume:Resume_afiq.pdf	2026-06-09 15:12:01
☐	Edit	☐	Copy	Delete	9 ApplicationReceived	jobId:1,candidateId:6,resume:Resume_afiq.pdf	2026-06-09 15:12:31
☐	Edit	☐	Copy	Delete	10 ApplicationReceived	jobId:1,candidateId:8,resume:Resume_aling.pdf	2026-06-13 22:06:21
☐	Edit	☐	Copy	Delete	11 ApplicationStatusUpdated	applicationId:6,status:SHORTLISTED,recruiterId:7	2026-06-13 22:12:12

The event_logs table records all published events including JobPosted, ApplicationReceived, ApplicationStatusUpdated, and InterviewScheduled, demonstrating the operation of the event bus.

		id	application_id	fit_score	keyword_matches	status		
☐	Edit	☐	Copy	Delete	6	6	40 Basic Criteria Met	REJECTED
☐	Edit	☐	Copy	Delete	7	7	85 Java, Software Architecture, SQL, Web Systems Mana...	SHORTLISTED

The screening_results table stores automated fit scores and screening outcomes generated by the ResumeScreeningConsumer service.

		id	application_id	schedule_time	meeting_link	feedback			
☐	Edit	☐	Copy	Delete	1	6	2026-06-15 13:17:00	https://webex.com	NULL

		id	candidate_id	message	created_at	is_read			
☐	Edit	☐	Copy	Delete	1	9	Interview scheduled on 2026-06-13T13. Meeting Link...	2026-06-15 13:38:14	0

The interviews and notifications tables verify successful execution of interview scheduling and notification services within the event-driven workflow.

5. Final Individual Reflection

This project helped me understand how Event-Driven Architecture (EDA) can be applied in a real-world recruitment system. I improved my skills in Java, JSP, MySQL, database integration, and system development. Through developing RecruitX, I also learned the importance of teamwork, problem-solving, and effective communication. Overall, this project provided valuable practical experience and strengthened my understanding of software architecture concepts.